Two-Timescale Optimization for E2E Network Slicing-Aided Cloud-Edge Collaborative Networks

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***Abstract*—To leverage the synergy between cloud computing (CC) and edge computing (EC) to support various services while reducing the CC/EC switching overhead, the two-timescale utility maximization problem for an end-to-end network slicing-aided cloud-edge collaborative network (E2E-CECN) is formulated covering both the high throughput and low delay service requirements. To solve the utility maximization problem while dynamically adjusting the weights of the E2E-CECN utility to accommodate the variation of users' service requests, we proposed a reward comparison double deep Q network algorithm to optimize the large timescale joint virtual basestation activation and CC-EC scheduling, and a reward comparison deep deterministic policy gradient algorithm to optimize the small timescale allocation of backhaul link capacity (BLC), CC/EC capability and transmission power. Numerical results show that both the high-throughput and low-delay service requirements can be satisfied simultaneously under the reasonable BLC usage and power consumption.**

***Index Terms*—Cloud-edge collaborative network, end-to-end network slicing, cloud computing, edge computing, two-timescale.**

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I. Introduction

With abundant computing resources, a cloud server (CS) is able to process a large amount of computation-intensive tasks. However, the growing data traffic in wireless networks may cause serious backhaul link (BL) congestion under the cloud computing (CC) paradigm. Although edge computing (EC) can alleviate the congestion of the BLs, the computing capability of an edge server (ES) is limited [1].

To tackle the above issues, the cloud-edge collaborative network (CECN) was proposed [2], [3], [4], [5]. Ren et al. [2] studied the joint communication and computing resource allocation for partial computation offloading in CECN to minimize the delay of end devices. To guarantee user fairness in cloud-edge collaboration enabled video service scenario, Wu et al. [3] proposed to make the video caching and resource allocation decisions at the cloud and edge, respectively, to maximize the minimum video coding rate for individual users. Lin et al. [4] maximized the secrecy energy efficiency for a cloud-satellite-base station (BS) collaborative network, where the cloud processing center links the satellite gateway and the BS together to manage the resources of the entire network. Laili et al. [5] investigated the CS-ES scheduling of the industrial Internet-of-things computation tasks to minimize the weighted sum of the task execution time and energy consumption. However, the optimizations for CECN in [2], [3], [4], [5] considered only the single service requirement. With the diversification of applications in the next generation wireless networks, CECNs can leverage end-to-end (E2E) network slicing to meet the differentiated service requirements of the users [16].

As a key technology of the next generation communication, E2E network slicing defines multiple logical E2E networks upon a common physical infrastructure to support various vertical applications. For the cloud-edge collaboration in network slicing, Wang et al. [6] studied the cloud-edge data centers collaborative deployment of the virtual network functions to optimize the delay, load balancing and bandwidth consumption for core network slicing. Tang et al. [7] investigated the cloud-edge multidimensional resources collaborative orchestration for resource slicing. In order to support diverse applications simultaneously with a low cost, it is necessary to explore the synergy between CECN and E2E network slicing, because it will enable the full exploitation of the distinct advantages of CC and EC while flexibly assembling the distributed communication and computing resources over the entire CECN.

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TABLE I
COMPARISON OF OUR WORK WITH THE EXISTING WORKS

Novelty	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[14]	Our Work
Studying the collaboration between Cloud and Edge	✓	✓		✓	✓	✓		✓	✓
Deploying network slices upon CECN					✓	✓			✓
Optimization for E2E Network slicing							✓		✓
Dynamic vBS activation							✓		✓
Dynamic CC-EC scheduling for each user	✓			✓				✓	✓
Allocation of BL resource	✓						✓		✓
Allocation of power resource		✓	✓			✓	✓	✓	✓
Allocation of CC and EC resources	✓					✓		✓	✓
Optimizing CC-EC scheduling and resource allocation in different timescales								✓	✓
Weighted sum of different objectives				✓	✓		✓		✓
Optimizing the performance indicators and costs simultaneously							✓		✓
Dynamic weight adjustments for different objectives									✓
DRL algorithm that compares the rewards under different weights									✓

To optimize the multiple objectives for E2E network slices, Wang et al. [8] defined the system-level utility as the weighted sum of coverage percentage, throughput, computing capability and delay. Similarly, the optimizations for CECN in [5] and for network slicing in [6] were based on the weighted sum of the different objectives. However, the weighted sum optimizations in the above works depended heavily on fixed weights, making them hard to adapt to the dynamicity of service requirements.

To address the above challenges, in this paper, we study an E2E network slicing-aided CECN (E2E-CECN), which contains two typical E2E network slices that provide the high-throughput and low-delay services, respectively. To accommodate the variation of users' service requirements, we define the E2E-CECN utility as a weighted sum of the E2E throughput and service delay with dynamic revenue weights, and formulate an optimization problem to maximize the utility by jointly optimizing the virtual base station activation (vBS-A), CC-EC scheduling, as well as the communication and computing resource allocation. To solve the formulated problem while keeping low overhead for switching CC/EC or BS on/off, a Reward Comparison Double Deep Q Network (RCDDQN) algorithm as well as a Reward Comparison Deep Deterministic Policy Gradient (RCDDPG) algorithm are proposed to optimize the large timescale vBS-A and CC-EC scheduling as well as the small timescale joint allocation of BL capacity (BLC), CC/EC capability and transmission power, respectively. Specially, the utility weight adjustments are determined together by the users' service requests and the target networks of both the RCDDQN and RCDDPG algorithms. The main contributions of our work are compared with the existing works in Table I.

In the following, we present our E2E-CECN model and define the system-level E2E-CECN utility in Sections II and III, respectively. The double-layered RCDDQN-RCDDPG algorithm is proposed in Section IV to maximize the E2E-CECN utility. In Section V, the simulation results are presented. Finally, we conclude the paper in Section VI.

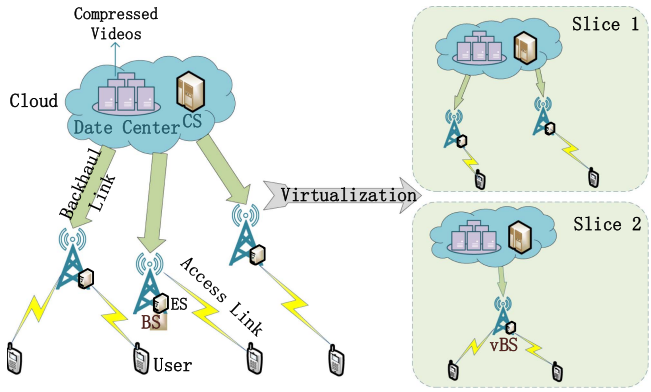


Fig. 1. E2E-CECN Model.

II. E2E-CECN MODEL

The E2E-CECN model shown in Fig. 1 is composed of a CS, N_B BSs each attached with an ES, and N_K users. Without loss of generality, we define two E2E network slices: Slice 1 for high throughput service, such as high-definition video [9], and slice 2 for low delay service, such as smooth video [10].

We first present the workflow of the E2E-CECN model exemplified by video services, but the model can be generalized to other scenarios such as Internet of vehicles and virtual/augmented reality. The compressed videos are stored in the cloud data center. Upon request by the users, the stored videos will undergo the data transmission and data decoding procedures. During data transmission, the requested videos will first be transmitted from the cloud to the BSs via the BLs, and then transmitted to the users over the wireless access links. The compressed videos will be decoded into the required definition version [11] either by the CS before the BL transmission or by the ESs before the access link transmission. Decoding a video by an ES saves the BLC cost for its BL transmission than decoding it by the CS [12]. The main variables and sets of this paper are listed in Table II.

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TABLE II
SUMMARY OF MAIN VARIABLES AND SETS

Notations	Descriptions
$\boldsymbol{N_K}$, $\boldsymbol{N_B}$	Sets of users, BSs in E2E-CECN
$\boldsymbol{o^s}(t)=\{o_k^s(t)\}$	Set of users' service request indicators of slice s at slot t
$\boldsymbol{a^s}(T)=\{a_b^s(T)\}$	Set of vBS-A indicators of slice s at epoch T
$\boldsymbol{N_K^s}(t)$, $\boldsymbol{N_B^s}(T)$	Set of users requesting the service of slice s at slot t, set of activated vBSs in slice s at slot t
H_B, P_B, F_B	Total BLC, total transmission power of one BS, total EC capability of one ES
$\boldsymbol{H^s}(t)=\{H_k^s(t)\}$	Set of BLC levels of slice s at slot t
$\boldsymbol{P^s}(t)=\{P_k^s(t)\}$	Set of transmission power levels of slice s at slot t
$\boldsymbol{F^s}(t)=\{\xi_k^s(t),f_k^s(t)\}$	Set of (CC and EC) computing capabilities of slice s at slot t
$\boldsymbol{\zeta^s}(T)=\{\zeta_k^s(T)\}$	Set of CC-EC scheduling indicators of slice s at epoch T
$\boldsymbol{h^s}(t)=\{h_{b,k}^s(t)\}$	Set of downlink channel power losses of slice s at slot t
$r_{b,k}^s$	Distance from vBS b to user k in slice s
$g_{b,k}^s(t)$	Whether user k in slice s is associated with vBS b at slot t
$\boldsymbol{Ad^s}(t)=\{Ad_k^s(t)\}$	Set of random data arrivals of the users in slice s at slot t
$R_\Sigma^s(t)$, $D_\Gamma^s(t)$	Total E2E throughput, user-specific average service delay of slice s at slot t
$H_\Sigma(t)$, $P_\Sigma(t)$	Total BLC, total transmission power of physical E2E-CECN at slot t

The small timescale is modeled by time slots each with the duration of ϵ second. The t-th slot is the time interval $[t\epsilon,(t+1)\epsilon)$, where $t\in\{0,1,2,\ldots\}$. The large timescale is represented by epochs each with the duration of Υ consecutive slots. The T-th epoch is the time interval $[T\Upsilon\epsilon,(T+1)\Upsilon\epsilon)$, where $T\in\{0,1,2,\ldots\}$.

The N_K users form the set of $\boldsymbol{N_K}$. The sets of users' service request indicators at slot t, $\boldsymbol{o^s}(t)=\{o_k^s(t)\}$ $(s\in\{1,2\},\ k\in\boldsymbol{N_K})$, are defined such that $o_k^s(t)=1$ if user k requests the service of slice s at slot t and $o_k^s(t)=0$ otherwise. All the users requesting the service of slice s at slot t form the set of users in slice s at slot t, denoted by $\boldsymbol{N_K^s}(t)$ with the size of $N_K^s(t)=\sum_{k\in\boldsymbol{N_K}}o_k^s(t)$.

The N_B BSs distributed on the ground form the set of $\boldsymbol{N_B}$. Each BS is mapped to two virtual BSs (vBSs) to serve the users in the two slices, respectively. Each vBS in slice s $(s\in\{1,2\})$ is activated according to its nearby users' service requests, and the two vBSs mapped from the same BS can be activated simultaneously to serve the two slices. To avoid frequently switching on/off vBSs [13], the vBS-A mode is adjusted once per epoch according to the binary vBS-A indicator sets $\boldsymbol{a^s}(T)=\{a_b^s(T)\}$ $(s\in\{1,2\},b\in\boldsymbol{N_B})$, where $a_b^s(T)=1$ if vBS b in slice s is activated at epoch T otherwise $a_b^s(T)=0$. The activated vBSs in slice s at epoch T form the set $\boldsymbol{N_B^s}(T)$ with the size of $N_B^s(T)=\sum_{b\in\boldsymbol{N_B}}a_b^s(T)$.

The total transmission power of a BS is P_B. Each BS connects to the CS via a BL with the capacity of H_B. We denote the BLC and transmission power allocated to user k in slice s at slot t by $H_k^s(t)$ and $P_k^s(t)$, respectively. The sets of BLC and transmission power levels of slice s at slot t are expressed as $\boldsymbol{H^s}(t)=\{H_k^s(t)\}$ and $\boldsymbol{P^s}(t)=\{P_k^s(t)\}$ $(s\in\{1,2\},k\in\boldsymbol{N_K^s}(t))$, respectively.

We assume that the CC capability of the CS is infinite and the EC capability of each ES is F_B. Let $\xi_k^s(t)$ and $f_k^s(t)$ denote the CC capability and EC capability allocated to user k in slice s at slot t, respectively, where $\xi_k^s(t)f_k^s(t)=0$. The computing capability sets of slice s at slot t are given by $\boldsymbol{f^s}(t)=\{\xi_k^s(t),f_k^s(t)\}$ $(s\in\{1,2\},k\in\boldsymbol{N_K^s}(t))$.

To avoid excessive CC/EC switching overhead [14], the requested data of a user will be processed by either the CS or an ES within one epoch. The sets of CC-EC scheduling indicators are defined as $\boldsymbol{\zeta^s}(T)=\{\zeta_k^s(T)\}$ $(s\in\{1,2\},k\in\boldsymbol{N_K^s}(t))$, where $\zeta_k^s(T)=1$ if the requested data of user k in slice s is processed by the CS at epoch T, and $\zeta_k^s(T)=0$ if the requested data of user k in slice s is processed by the ESs at epoch T.

III. UTILITY DEFINITION FOR E2E-CECN

Each user is associated with its nearest vBS, and the down-link channels of the BSs exhibit independent Rayleigh fading. The orthogonal frequency division multiple access method is used in CECN. Hence, the signal-to-interference plus noise ratio from vBS b to user k in slice s at slot t is given by

$$SINR_k^s(t)=\frac{P_k^s(t)h_{b,k}^s(t)g_{b,k}^s(t)\left(r_{b,k}^s\right)^{-\alpha}}{\sum\limits_{j\in\boldsymbol{N_K^s}(t)\backslash\{k\}}\ \sum\limits_{d\in\boldsymbol{N_B^s}(T)\backslash\{b\}}P_j^s(t)h_{d,k}^s(t)g_{d,j}^s(t)\left(r_{d,k}^s\right)^{-\alpha}+\sigma^2},\tag{1}$$

where $t\in[T\Upsilon,(T+1)\Upsilon)$; $h_{b,k}^s(t)$ is the downlink channel power loss from vBS b to user k in slice s at slot t and follows an independent exponential distribution with unit mean; $r_{b,k}^s$ is the distance from vBS b to user k in slice s; α and σ^2 are the path loss exponent and the power of the additive white Gaussian noise, respectively. The binary vBS association indicator $g_{b,k}^s(t)$ is defined such that $g_{b,k}^s(t)=1$ if user k in slice s is associated with vBS b at slot t where $b=\operatorname{argmin}_{b'\in\boldsymbol{N_B^s}(t)}r_{b',k}^s$, and otherwise $g_{b,k}^s(t)=0$. The set of the downlink channel power losses of slice s at slot t is denoted by $\boldsymbol{h^s}(t)=\{h_{b,k}^s(t)\}$ $(b\in\boldsymbol{N_B^s}(T),k\in\boldsymbol{N_K^s}(t),s\in\{1,2\})$. Hence, the downlink rate of

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user k in slice s at slot t is obtained as

$$R_k^s(t) = B_K \log_2 \left[1 + SINR_k^s(t)\right]. \tag{2}$$

where B_K is the downlink bandwidth of each user. The total E2E throughput of slice s at slot t is given by [15]

$$R_\Sigma^s(t) = \sum_{k \in \boldsymbol{N_K^s}(t)} \min\left\{R_k^s(t), H_k^s(t)\right\}. \tag{3}$$

We denote the sets of the random data arrivals of the users on the CS (if $\zeta_k^s(T) = 1$) or the BL (if $\zeta_k^s(T) = 0$) in slice s at slot t as $\boldsymbol{Ad^s}(t) = \{Ad_k^s(t)\}$ $(s \in \{1,2\}, k \in \boldsymbol{N_K^s}(t))$, which is independent and identically distributed over the time slots and follows a Poisson arrival process with the average rate of γ_s^a (in bits/s). If the requested data of user k in slice s is processed by the CS at epoch T, the service delay of user k contains the CC delay, BL transmission delay and access link transmission delay as

$$G_k^s(t) = \frac{c_k^s Ad_k^s(t)}{\xi_k^s(t)} + \frac{n_k^s Ad_k^s(t)}{H_k^s(t)} + \frac{n_k^s Ad_k^s(t)}{R_k^s(t)}, \tag{4}$$

where c_k^s (in CPU cycle/bit) is the required CPU cycles for processing each bit of the data requested by user k in slice s, and n_k^s is the ratio between the sizes of data before processing and data after processing requested by user k in slice s [12].

If the requested data of user k in slice s is processed by the ES at epoch T, the service delay of user k consists of the BL transmission delay, EC delay and access link transmission delay as follows

$$Z_k^s(t) = \frac{Ad_k^s(t)}{H_k^s(t)} + \frac{c_k^s Ad_k^s(t)}{f_k^s(t)} + \frac{n_k^s Ad_k^s(t)}{R_k^s(t)}. \tag{5}$$

The total service delay of slice s at slot t is

$$D_\Sigma^s(t) = \sum_{k \in \boldsymbol{N_K^s}(t)} \left[\zeta_k^s(T) G_k^s(t) + (1 - \zeta_k^s(T))\, Z_k^s(t)\right]. \tag{6}$$

and the user-specific average service delay of slice s is

$$D_\Gamma^s(t) = \frac{D_\Sigma^s(t)}{N_K^s(t)}. \tag{7}$$

The costs of the physical CECN include the total BLC and total power consumption. Therein, the total BLC of the physical CECN at slot t is

$$H_\Sigma(t) = \sum_{s \in \{1,2\}} \sum_{k \in \boldsymbol{N_K^s}(t)} H_k^s(t). \tag{8}$$

The total power consumption of the physical CECN contains the transmission power, CC power (i.e., $\kappa[\xi_k^s(t)]^3$[8] [12]) and EC power (i.e., $\kappa[f_k^s(t)]^3$), which is

$$P_\Sigma(t) = \sum_{s \in \{1,2\}} \sum_{k \in \boldsymbol{N_K^s}(t)} \Big\{P_k^s(t) + \zeta_k^s(T)\kappa\left[\xi_k^s(t)\right]^3 + (1 - \zeta_k^s(T))\,\kappa\left[f_k^s(t)\right]^3\Big\}. \tag{9}$$

The utility of E2E-CECN is defined as the sum-gain of the two E2E network slices from the total E2E throughput and total service delay, respectively, minus the total costs, i.e.,

$$U(t) = m_1(t) R_\Sigma^1(t) + \left[\psi N_K^2(t) - m_2(t) D_\Gamma^2(t)\right] - \eta \cdot \max\left\{D_\Gamma^2(t) - \rho, 0\right\} - \left\{\omega H_\Sigma(t) + \beta P_\Sigma(t)\right\}, \tag{10}$$

where the dynamic weights $m_1(t)$ and $m_2(t)$ are the revenues per unit throughput and per unit delay at slot t, respectively; ω and β are the costs per unit BLC and per unit power, respectively; ψ is an initial delay gain coefficient to ensure the non-negativity of the utility. To guarantee the delay requirements of the users in slice 2, a penalty term $\eta \cdot \max\{D_\Gamma^2(t) - \rho, 0\}$ is introduced such that the penalty will be paid if the average service delay of slice 2 is higher than a threshold ρ, where η is a positive penalty coefficient [8].

IV. Two-Timescale Utility Maximization

A. Two-Timescale Utility Maximization Problem Formulation

The E2E-CECN optimization is formulated as a two-timescale utility maximization problem $\mathcal{P}^*$ as

$$\begin{aligned}
\mathcal{P}^*: \quad & \min_{\{\boldsymbol{a^s}(T), \boldsymbol{\zeta^s}(T), \boldsymbol{H^s}(t), \boldsymbol{P^s}(t), \boldsymbol{f^s}(t)\}} -U(t) \\
\text{s.t.} \quad & C1: a_b^s(T), \zeta_k^s(T) \in \{0,1\}, \\
& \quad \forall s \in \{1,2\}, b \in \boldsymbol{N_B^s}(T), k \in \boldsymbol{N_K^s}(t), \\
& C2: \sum_{s \in \{1,2\}} \sum_{k \in \boldsymbol{N_K^s}(t)} H_k^s(t) g_{b,k}^s(t) \le H_B, \forall b \in \boldsymbol{N_B^s}(T), \\
& C3: \sum_{s \in \{1,2\}} \sum_{k \in \boldsymbol{N_K^s}(t)} P_k^s(t) g_{b,k}^s(t) \le P_B, \forall b \in \boldsymbol{N_B^s}(T), \\
& C4: \sum_{s \in \{1,2\}} \sum_{k \in \boldsymbol{N_K^s}(t)} \zeta_k^s(T) f_k^s(t) g_{b,k}^s(t) \le F_B, \forall b \in \boldsymbol{N_B^s}(T), \\
& C5: D_\Gamma^2(t) < \rho,
\end{aligned} \tag{11}$$

In the constraints, $C1$ specifies the value ranges of the binary vBS-A and CC-EC scheduling indicators; Constraints $C2$, $C3$ and $C4$ guarantee that the BLC, transmission power and EC capability allocated to the users associated with the same BS would not exceed the total BLC, total transmission power and total EC capability of one BS, respectively; $C5$ is the average service delay constraint of slice 2.

The optimization problem $\mathcal{P}^*$ is an NP-hard, non-convex problem (The proofs for the NP-hardness and non-convexity of $\mathcal{P}^*$ will be shown in Appendix A and Appendix B, respectively), making it difficult to solve using traditional convex optimization methods. Therefore, the deep reinforcement learning (DRL) algorithm is used to solve it. Meanwhile, the problem contains both the continuous and discrete variables, which vary in different timescales and make it hard to solve $\mathcal{P}^*$ directly. Hence, we decouple the optimization problem $\mathcal{P}^*$ into the large timescale joint vBS-A and CC-EC scheduling subproblem $\mathcal{P}1$ and the small timescale joint communication and computing resources allocation subproblem $\mathcal{P}2$ as

$$\begin{aligned}
\mathcal{P}1: \quad & \min_{\{\boldsymbol{a^s}(T), \boldsymbol{\zeta^s}(T)\}} -U(T), \\
\text{s.t.} \quad & C1,
\end{aligned} \tag{12}$$

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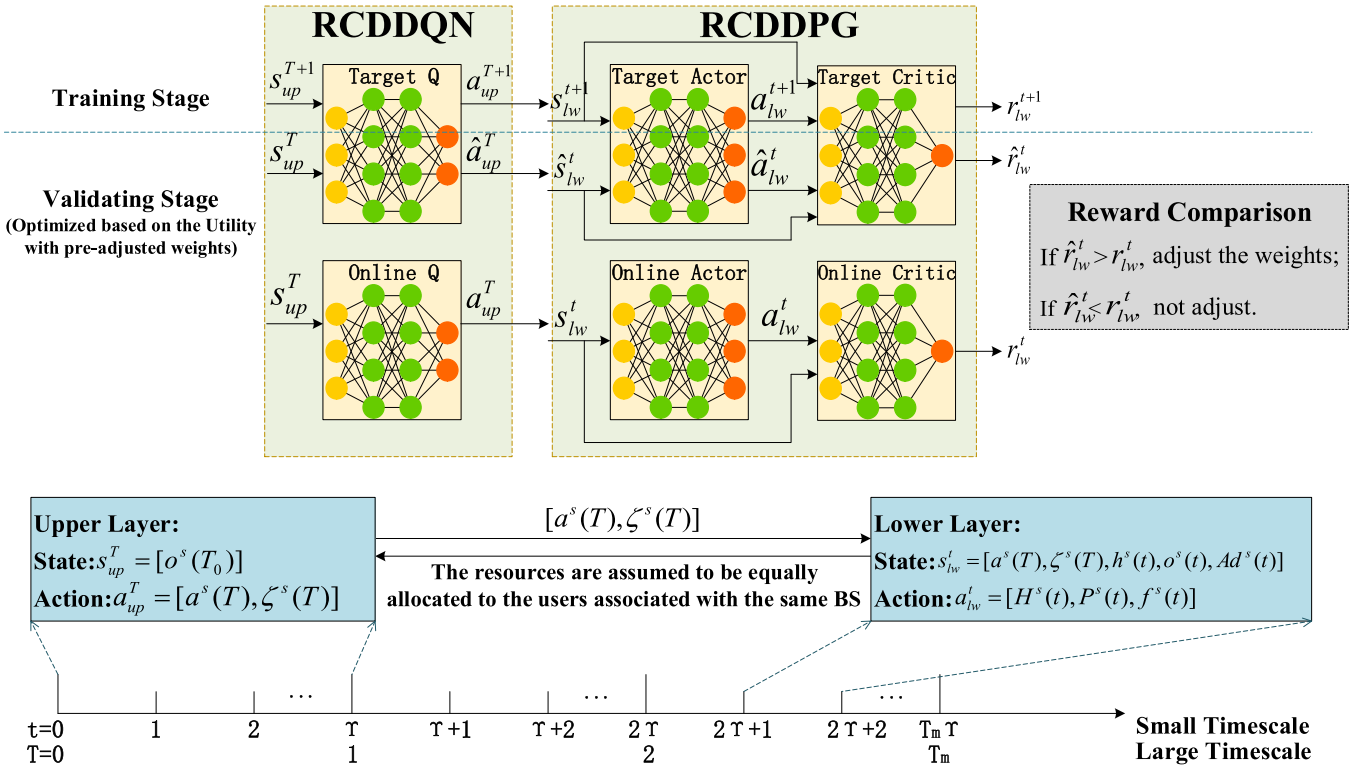


Fig. 2. RCDDQN-RCDDPG Algorithm.

and

$$\mathcal{P}2: \min_{\{\boldsymbol{H}^s(t),\boldsymbol{P}^s(t),\boldsymbol{f}^s(t)\}} -U(t),$$
$$\text{s.t.} \quad C2-C4, C5. \tag{13}$$

respectively.

B. Double-Layered DRL-Based Utility Maximization

We propose a double-layered RCDDQN-RCDDPG algorithm as shown in Fig. 2 to solve the two-timescale utility maximization problem. Specifically, the RCDDQN algorithm is used to solve subproblem $\mathcal{P}1$ with the discrete variables $\boldsymbol{a}^s(T)$ and $\boldsymbol{\zeta}^s(T)$ in the upper layer, while the RCDDPG algorithm is employed to solve subproblem $\mathcal{P}2$ with the continuous variables $\boldsymbol{H}^s(t)$, $\boldsymbol{P}^s(t)$ and $\boldsymbol{f}^s(t)$ in the lower layer. Our algorithm contains the training stage and validating stage. The basic revenue weight m_s^0 of slice s is used during the training stage of our algorithm and also used as the initial weight of the validating stage.

1) Upper Layer: RCDDQN-Based Joint vBS-A and CC-EC Scheduling: The RCDDQN algorithm is used to obtain the optimal vBS-A and CC-EC scheduling at the large timescale.

State Space: The state space of the upper layer is the users' service request states at the initial of the current epoch T_0:

$$\boldsymbol{s}_{\boldsymbol{up}}^{\boldsymbol{T}} = [\boldsymbol{o}^s(T_0)]. \tag{14}$$

Action Space: The action space of the upper layer contains the vBS-A and CC-EC scheduling indicators:

$$\boldsymbol{a}_{\boldsymbol{up}}^{\boldsymbol{T}} = [\boldsymbol{a}^s(T), \boldsymbol{\zeta}^s(T)], \tag{15}$$

Reward Function: The reward function of the upper layer is the utility of E2E-CECN determined by the state and action of the upper layer at epoch T:

$$r_{up}^T = U\left(\boldsymbol{s}_{\boldsymbol{up}}^{\boldsymbol{T}}, \boldsymbol{a}_{\boldsymbol{up}}^{\boldsymbol{T}}\right). \tag{16}$$

The Deep Neural Network of the RCDDQN Algorithm: The RCDDQN algorithm contains one online Q network and one target Q network with the parameters of θ_{up}^Q and $\theta_{up}^{Q'}$, respectively, and the state-action value function of the upper layer is denoted by $Q_{up}(\boldsymbol{s}_{\boldsymbol{up}}^{\boldsymbol{T}}, \boldsymbol{a}_{\boldsymbol{up}}^{\boldsymbol{T}}|\theta_{up}^Q)$. We aim to minimize the loss function of the upper layer to update the online Q network parameter θ_{up}^Q during the training stage, and the loss function is given by

$$L_{up} = E\left[\left(r_{up}^T + \gamma_{up} Q'_{up}\left(\boldsymbol{s}_{\boldsymbol{up}}^{\boldsymbol{T+1}}, \underset{\boldsymbol{a}_{\boldsymbol{up}}}{\operatorname{argmax}} Q_{up}\left(\boldsymbol{s}_{\boldsymbol{up}}^{\boldsymbol{T+1}}, \boldsymbol{a}_{\boldsymbol{up}}^{\boldsymbol{T+1}}|\theta_{up}^Q\right)|\theta_{up}^{Q'}\right) - Q_{up}\left(\boldsymbol{s}_{\boldsymbol{up}}^{\boldsymbol{T}}, \boldsymbol{a}_{\boldsymbol{up}}^{\boldsymbol{T}}|\theta_{up}^Q\right)\right)^2\right], \tag{17}$$

where $\gamma_{up} \in [0,1]$ is a discount factor, and $\boldsymbol{a}_{\boldsymbol{up}}$ is the set of candidate upper layer actions; $\boldsymbol{s}_{\boldsymbol{up}}^{\boldsymbol{T+1}}$ and $\boldsymbol{a}_{\boldsymbol{up}}^{\boldsymbol{T+1}}$ are the state and action of the upper layer at the next epoch, respectively. The target Q network parameter is updated according to the online Q network parameter, i.e., $\theta_{up}^{Q'} = \theta_{up}^Q$, every T_e epochs.

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Algorithm 1: Upper Layer: RCDDQN-based Joint vBS-A and CC-EC Scheduling.

Initialization:
Initialize θ_{up}^{Q};
Initialize $\theta_{up}^{Q'} \leftarrow \theta_{up}^{Q}$;
Obtain the initial state space of the upper layer $\boldsymbol{s_{up}^{0}}$;
Initialize the replay buffer of the upper layer $\boldsymbol{D_{up}}$;
For $T = 1, 2, \cdots T_m$ **do**
  Random select action $\boldsymbol{a_{up}^{T}}$ with the exploring probability of υ, otherwise $\boldsymbol{a_{up}^{T}} = \text{argmax}_{\boldsymbol{a_{up}}} Q_{up}(\boldsymbol{s_{up}^{T}}, \boldsymbol{a_{up}^{T}}|\theta_{up}^{Q})$;
  Take $\boldsymbol{a_{up}^{T}}$, obtain the reward r_{up}^{T} and the next state $\boldsymbol{s_{up}^{T+1}}$;
  Store the data tuple $(\boldsymbol{s_{up}^{T}}, \boldsymbol{a_{up}^{T}}, r_{up}^{T}, \boldsymbol{s_{up}^{T+1}})$ in $\boldsymbol{D_{up}}$;
  If Replay buffer $\boldsymbol{D_{up}}$ is full **do**
    Select D_{up}^{mn} mini-batches randomly from $\boldsymbol{D_{up}}$;
    Update θ_{up}^{Q} by the gradient descent method to minimize the loss function $\nabla_{\theta_{up}^{Q}} L_{up} \approx \nabla_{\theta_{up}^{Q}} E[(r_{up}^{T} + \gamma_{up} Q'_{up}(\boldsymbol{s_{up}^{T+1}}, \text{argmax}_{\boldsymbol{a_{up}}} Q_{up}(\boldsymbol{s_{up}^{T+1}}, \boldsymbol{a_{up}^{T+1}}|\theta_{up}^{Q})|\theta_{up}^{Q'}) - Q_{up}(\boldsymbol{s_{up}^{T}}, \boldsymbol{a_{up}^{T}}|\theta_{up}^{Q})) \cdot \nabla_{\theta_{up}^{Q}} Q(\boldsymbol{s_{up}^{T}}, \boldsymbol{a_{up}^{T}}|\theta_{up}^{Q})]$;
    If $T\%T_e = 0$ **do**
      Update $\theta_{up}^{Q'} \leftarrow \theta_{up}^{Q}$;
    End If
  End If
  Update the state $\boldsymbol{s_{up}^{T}} \leftarrow \boldsymbol{s_{up}^{T+1}}$;
End For

2) Lower Layer: RCDDPG-Based Joint Communication and Computing Resources Allocation: The RCDDPG algorithm is employed to determine the optimal communication and computing resources allocation at the small timescale.

State Space: The state space of the lower layer contains the vBS-A and CC-EC scheduling strategy obtained by the upper layer, the downlink channel power losses, users' service request states and data arrivals at slot t as

$$\boldsymbol{s_{lw}^{t}} = [\boldsymbol{a^{s}}(T), \boldsymbol{\zeta^{s}}(T), \boldsymbol{h^{s}}(t), \boldsymbol{o^{s}}(t), \boldsymbol{Ad^{s}}(t)], \quad (18)$$

Action Space: The action space of the lower layer includes the allocations of the BLC, transmission power and CC/EC capability at slot t as

$$\boldsymbol{a_{lw}^{t}} = [\boldsymbol{H^{s}}(t), \boldsymbol{P^{s}}(t), \boldsymbol{f^{s}}(t)], \quad (19)$$

If some variables in $\boldsymbol{a_{lw}^{t}}$ violate the constraints $C2$-$C4$, the violative variables will be modified to the lower values.

Reward Function: The reward function of the lower layer is the utility of E2E-CECN determined by the state and action of the lower layer at the current slot, denoted as

$$r_{lw}^{t} = U(\boldsymbol{s_{lw}^{t}}, \boldsymbol{a_{lw}^{t}}). \quad (20)$$

The Deep Neural Network of the RCDDPG Algorithm: The parameters of the online actor network, online critic network, target actor network and target critic network in the RCDDPG algorithm are denoted by θ_{lw}^{μ}, $\theta_{lw}^{\mu'}$, θ_{lw}^{Q} and $\theta_{lw}^{Q'}$, respectively. We have the policy function of the lower layer as $\mu(\boldsymbol{s_{lw}^{t}}|\theta_{lw}^{\mu})$ for mapping the current state to a specific action, and the state-action value function is $Q_{lw}(\boldsymbol{s_{lw}^{t}}\boldsymbol{a_{lw}^{t}}|\theta_{lw}^{Q})$.

Algorithm 2: Lower Layer: RCDDPG-based Joint Communication and Computing Resource Allocation.

Initialization:
Initialize θ_{lw}^{Q} and θ_{lw}^{μ};
Initialize $\theta_{lw}^{Q'}$ and $\theta_{lw}^{\mu'}$: $\theta_{lw}^{Q'} \leftarrow \theta_{lw}^{Q}$, $\theta_{lw}^{\mu'} \leftarrow \theta_{lw}^{\mu}$;
Initialize the replay buffer $\boldsymbol{D_{lw}}$;
For $T = 1, 2, \cdots T_m$ **do**
  Perform Step 3 of Algorithm 1 to generate the action of the upper layer, and obtain the initial state of the lower layer $\boldsymbol{s_{lw}^{0}}$;
  For $t = 1, 2, \cdots \Upsilon$ **do**
  Select action $\boldsymbol{a_{lw}^{t}} = \mu(\boldsymbol{s_{lw}^{t}}) + \mathcal{N}_0$ based on the current deterministic policy μ with the stochastic noise $\mathcal{N}_0$;
  Check whether the variables in action $\boldsymbol{a_{lw}^{t}}$ violate the constraints and modify the violative variables
    Take $\boldsymbol{a_{lw}^{t}}$, obtain the reward r_{lw}^{t} and the next state $\boldsymbol{s_{lw}^{t+1}}$;
    Store the data tuple $(\boldsymbol{s_{lw}^{t}}, \boldsymbol{a_{lw}^{t}}, r_{lw}^{t}, \boldsymbol{s_{lw}^{t+1}})$ in $\boldsymbol{D_{lw}}$;
    If Replay buffer $\boldsymbol{D_{lw}}$ is full **do**
      Select D_{lw}^{mn} mini-batches randomly from $\boldsymbol{D_{lw}}$;
    Update θ_{lw}^{Q} by the gradient descent method to minimize the loss function $\nabla_{\theta_{lw}^{Q}} L_{lw} \approx \nabla_{\theta_{lw}^{Q}} E[(r_{lw}^{t} + \gamma_{lw} Q'_{lw}(\boldsymbol{s_{lw}^{t+1}}, \mu(\boldsymbol{s_{lw}^{t+1}}|\theta_{lw}^{\mu'})|\theta_{lw}^{Q'}) - Q_{lw}(\boldsymbol{s_{lw}^{t}}, \boldsymbol{a_{lw}^{t}}|\theta_{lw}^{Q})) \cdot \nabla_{\theta_{lw}^{Q}} Q_{lw}(\boldsymbol{s_{lw}^{t}}, \boldsymbol{a_{lw}^{t}}|\theta_{lw}^{Q})]$;
    Update θ_{lw}^{μ} by the gradient descent method $\nabla_{\theta_{lw}^{\mu}} J \approx \nabla_{\theta_{lw}^{\mu}} E[\nabla_a Q_{lw}(\boldsymbol{s_{lw}^{t}}, \boldsymbol{a_{lw}^{t}}|\theta_{lw}^{Q}) \cdot \nabla_{\theta^{\mu}} \mu(\boldsymbol{s_{lw}^{t}}|\theta_{lw}^{\mu})]$;
      Soft update $\theta_{lw}^{Q'}$ and $\theta_{lw}^{\mu'}$;
      End If
    Update the state: $\boldsymbol{s_{lw}^{t}} \leftarrow \boldsymbol{s_{lw}^{t+1}}$;
  End For
End For

The expectation of the long-term discounted accumulative reward of the lower layer is denoted by

$$J = E\left[r_0 + \gamma_{lw} r_1 + (\gamma_{lw})^2 r_2 + \cdots + (\gamma_{lw})^n r_n\right], \quad (21)$$

where γ_{lw} is the discount factor of the lower layer, and r_0, r_1, r_2,..., r_n are the rewards of the lower layer at each slot.

The loss function of the online critic network is given by

$$L_{lw} = E\left[\left(r_{lw}^{t} + \gamma_{lw} Q'_{lw}\left(\boldsymbol{s_{lw}^{t+1}}, \mu\left(\boldsymbol{s_{lw}^{t+1}}|\theta_{lw}^{\mu'}\right)|\theta_{lw}^{Q'}\right) - Q_{lw}\left(\boldsymbol{s_{lw}^{t}}, \boldsymbol{a_{lw}^{t}}|\theta_{lw}^{Q}\right)\right)^2\right], \quad (22)$$

where $\boldsymbol{s_{lw}^{t+1}}$ and $\boldsymbol{a_{lw}^{t+1}}$ are the state and action of the lower layer at slot $t+1$, respectively. Soft update method is used to update the target actor and critic network parameters [8].

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3) Rewards Comparison and Weights Adjustment: During the validating stage, the revenue weights are pre-adjusted according to the variation of the users' service requests. Specifically, the pre-adjusted revenue weight $m'_s(t)$ $(s = 1, 2)$ of slice s is scaled by a factor of $(1 + \delta)$ at slot t compared with the previous revenue weight, i.e., $m'_s(t) = (1 + \delta)m_s(t - 1)$, where $0 < \delta < 1$ if the number of users requesting the service of slice s increases (i.e., $N^s_K(t) > N^s_K(t - 1)$), otherwise $-1 < \delta < 0$. Based on the utility with the pre-adjusted weights, the target Q network of the RCDDQN algorithm is used to obtain the contrast upper layer action $\hat{\boldsymbol{a}}^{\boldsymbol{T}}_{\boldsymbol{up}}$ at the initial of each epoch to generate the contrast lower layer state $\hat{\boldsymbol{s}}^{\boldsymbol{t}}_{\boldsymbol{lw}}$, and the target actor network and target critic network of the RCDDPG algorithm are used to obtain the contrast lower layer action $\hat{\boldsymbol{a}}^{\boldsymbol{t}}_{\boldsymbol{lw}}$ and the contrast reward $\hat{r}^t_{lw} = U(\hat{\boldsymbol{s}}^{\boldsymbol{t}}_{\boldsymbol{lw}}, \hat{\boldsymbol{a}}^{\boldsymbol{t}}_{\boldsymbol{lw}})$ at each slot, respectively. Meanwhile, the online networks of both the RCDDQN and RCDDPG algorithms run based on the utility with the un-adjusted weights. Then, the contrast reward will be compared with the reward r^t_{lw} obtained by the online networks to deter-mine whether or not to adjust the revenue weights, i.e., $m_s(t) = m'_s(t)$ if $\hat{r}^t_{lw} > r^t_{lw}$, and $m_s(t)$ keeps unchanged otherwise. The action with the greater reward will be taken.

The training stages of the lower layer and upper layer of our algorithm are shown in Algorithm 1 and Algorithm 2, respectively. The validating stage of our algorithm is shown in Algorithm 3.

4) Complexity Analysis of the Algorithm: Based on the deep neural network model, we will analyze the computational complexity of the double-layered RCDDQN-RCDDPG algorithm. The input layers, output layers and hidden layers in the deep Q network of the RCDDQN algorithm as well as the actor and critic networks of the RCDDPG algorithm are all fully connected. We assume that the RCDDQN-RCDDPG algorithm will converge after V episodes and E time epochs.

For the RCDDQN algorithm in the upper layer, the deep Q network takes the users' service request states as input, and outputs the rewards under different possible values of the vBS-A and CC-EC scheduling indicators. We denote the dimensions of the input and output of the deep Q network as $|\mathcal{S}_{up}|$ and $|\mathcal{A}_{up}|$, respectively. Besides, the number of layers of the deep Q network is denoted by N_Q, and the number of neurons in the l-th layer of the deep Q network is denoted by n^l_Q. Hence, we have the complexity of the RCDDQN algorithm as

$$\mathcal{O}\left(VE\left(n^1_Q|\mathcal{S}_{up}| + \sum_{l=1}^{N_Q-1} n^l_Q n^{l+1}_Q + n^{N_Q}_Q|\mathcal{A}_{up}|\right)\right). \quad (23)$$

For the RCDDPG algorithm, the actor network takes the state space of the lower layer as input and outputs the action space, while the critic network takes both the state and action spaces of the lower layer as input and outputs the reward. The dimensions of the lower layer state and action spaces are denoted by $|\mathcal{S}_{lw}|$ and $|\mathcal{A}_{lw}|$, respectively. Then, the dimensions of the input and output of the actor network are $|\mathcal{S}_{lw}|$ and $|\mathcal{A}_{lw}|$, respectively; the dimensions of the input and output of the critic network are $|\mathcal{S}_{lw}| + |\mathcal{A}_{lw}|$ and 1, respectively. We denote the numbers of layers of the actor and critic networks by N_A and N_C, respectively; the numbers of neurons in the l-th layer of the actor and critic networks are n^l_A and n^l_C, respectively. We have the complexity of the RCDDPG algorithm as

$$\mathcal{O}\left(VE\Upsilon\left(\left(n^1_A|\mathcal{S}_{lw}| + \sum_{l=1}^{N_A-1} n^l_A n^{l+1}_A + n^{N_A}_A|\mathcal{A}_{lw}|\right) + \left(n^1_C\left(|\mathcal{S}_{lw}| + |\mathcal{A}_{lw}|\right) + \sum_{l=1}^{N_C-1} n^l_C n^{l+1}_C + n^{N_C}_C\right)\right)\right). \quad (24)$$

Algorithm 3: Validating Stage of the RCDDQN-RCDDPG Algorithm.

Initialize the revenue weights: $m_1(0) \leftarrow m^0_1$, $m_2(0) \leftarrow m^0_2$;
Obtain the initial state space of the upper layer $\boldsymbol{s}^{\boldsymbol{0}}_{\boldsymbol{up}}$;
For $T = 0, 1, 2, \ldots T_m$ **do**
  Select action $\boldsymbol{a}^{\boldsymbol{T}}_{\boldsymbol{up}}$ by the online Q network, and obtain the initial state space of the lower layer $\boldsymbol{s}^{\boldsymbol{0}}_{\boldsymbol{lw}}$;
  For $t = 0, 1, 2, \ldots \Upsilon$ **do**
    Select lower layer action $\boldsymbol{a}^{\boldsymbol{t}}_{\boldsymbol{lw}}$ and obtain the reward r^t_{lw};
    If $T > 0$ or $t > 0$ **do**
      Pre-adjust the avenue weights: $m'_s(t) = (1 \pm \delta)m_s(t - 1)$ if $N^s_K(t) \neq N^s_K(t - 1)$;
      If $t = 0$ **do**
        Select contrast action $\hat{\boldsymbol{a}}^{\boldsymbol{T}}_{\boldsymbol{up}}$ by the target Q network, and obtain the initial contrast state space of the lower layer $\hat{\boldsymbol{s}}^{\boldsymbol{0}}_{\boldsymbol{lw}}$;
        Select lower layer contrast action $\hat{\boldsymbol{a}}^{\boldsymbol{0}}_{\boldsymbol{lw}}$ and obtain the contrast reward $\hat{r}^0_{lw}$;
        If $\hat{r}^0_{lw} > r^0_{lw}$ **do**
          $m_s(t) = m'_s(t)$
          Action $\hat{\boldsymbol{a}}^{\boldsymbol{T}}_{\boldsymbol{up}}$ is taken to obtain the lower layer state space $\boldsymbol{s}^{\boldsymbol{t}}_{\boldsymbol{lw}}$ and the next upper layer state $\boldsymbol{s}^{\boldsymbol{T+1}}_{\boldsymbol{up}}$
          Action $\hat{\boldsymbol{a}}^{\boldsymbol{0}}_{\boldsymbol{lw}}$ is taken to obtain the reward r^t_{lw} and the next lower layer state $\boldsymbol{s}^{\boldsymbol{t+1}}_{\boldsymbol{lw}}$
        Else do
          Action $\boldsymbol{a}^{\boldsymbol{T}}_{\boldsymbol{up}}$ is taken to obtain the lower layer state space $\boldsymbol{s}^{\boldsymbol{t}}_{\boldsymbol{lw}}$ and the next upper layer state $\boldsymbol{s}^{\boldsymbol{T+1}}_{\boldsymbol{up}}$
          Action $\boldsymbol{a}^{\boldsymbol{0}}_{\boldsymbol{lw}}$ is taken to obtain the reward r^t_{lw} and the next lower layer state $\boldsymbol{s}^{\boldsymbol{t+1}}_{\boldsymbol{lw}}$
        End If
      Else If
        Select lower layer contrast action $\hat{\boldsymbol{a}}^{\boldsymbol{t}}_{\boldsymbol{lw}}$ and obtain the contrast reward $\hat{r}^t_{lw}$;
        If $\hat{r}^t_{lw} > r^t_{lw}$ **do**
          $m_s(t) = m'_s(t)$
          Action $\hat{\boldsymbol{a}}^{\boldsymbol{t}}_{\boldsymbol{lw}}$ is taken to obtain the reward r^t_{lw} and the next state $\boldsymbol{s}^{\boldsymbol{t+1}}_{\boldsymbol{lw}}$
        Else do
          Action $\boldsymbol{a}^{\boldsymbol{t}}_{\boldsymbol{lw}}$ is taken to obtain the reward r^t_{lw} and the next state $\boldsymbol{s}^{\boldsymbol{t+1}}_{\boldsymbol{lw}}$
        End If
      End If
    End If
    Update the lower layer state $\boldsymbol{s}^{\boldsymbol{t}}_{\boldsymbol{lw}} \leftarrow \boldsymbol{s}^{\boldsymbol{t+1}}_{\boldsymbol{lw}}$;
  End For
  Update the upper layer state $\boldsymbol{s}^{\boldsymbol{T}}_{\boldsymbol{up}} \leftarrow \boldsymbol{s}^{\boldsymbol{T+1}}_{\boldsymbol{up}}$;
End For

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Therefore, the complexity of the RCDDQN-RCDDPG algorithm is given by

$$\mathcal{O}\left(VE\left(n_Q^1|\mathcal{S}_{up}|+\sum_{l=1}^{N_Q-1}n_Q^l n_Q^{l+1}+n_Q^{N_Q}|\mathcal{A}_{up}|\right)+VE\Upsilon\left(\left(n_A^1|\mathcal{S}_{lw}|+\sum_{l=1}^{N_A-1}n_A^l n_A^{l+1}+n_A^{N_A}|\mathcal{A}_{lw}|\right)+\left(n_C^1\left(|\mathcal{S}_{lw}|+|\mathcal{A}_{lw}|\right)+\sum_{l=1}^{N_C-1}n_C^l n_C^{l+1}+n_C^{N_C}\right)\right)\right). \quad (25)$$

V. Numerical Results and Discussions

In this section, the numerical results are presented to verify the effectiveness of the proposed E2E-CECN in conjunction with the double-layered RCDDQN-RCDDPG algorithm. According to [8], [11], [12] and [17], the parameters are set as follows: $N_B = 4$, $H_B = 500$ kbps, $B_K = 50$ MHz, $P_B = 30$ W, $F_B = 20$ M CPU cycle/s, $\kappa = 10^{-26}(\mathrm{W}\cdot s^3/\mathrm{cycle}^3)$, $\sigma = 1.591$, $\epsilon = 50$ ms, $\Upsilon = 2000$, $m_1^0 = 0.001$(s/bit), $m_2^0 = 1000$(/s), $\psi = 5000$, $\eta = 0.3$, $\rho = 0.5$ s, $\omega = 0.0002$(s/bit) and $\beta = 50$(/W).

To manifest the superiority of the E2E-CECN model, E2E network slicing-aided CC network (E2E-CCN) and E2E network slicing-aided EC network (E2E-ECN) are used as the benchmarks for comparison. Meanwhile, we compare the distinct epoch lengths (i.e., $\Upsilon = 1000$ and $\Upsilon = 4000$) for the RCDDQN-RCDDPG algorithm with the epoch length $\Upsilon = 2000$ to justify our choice of epoch length. To verify the advantage of the proposed RCDDQN-RCDDPG algorithm in solving the two timescale optimization problem, we compare it with the existing algorithms: 1) Double Deep Q Network (DDQN)-Deep Deterministic Policy Gradient (DDPG) algorithm [18]: the revenue weights of the utility are fixed [6] [8], and the DDQN algorithm is used to optimize the large timescale joint vBS-A and CC-EC scheduling, while the DDPG algorithm is used to optimize the small timescale joint allocation of communication and computing resources. 2) Twin-Actor DDPG algorithm [19], [20]: two online actor networks are included in the DDPG framework to solve the large timescale optimization problem and the small timescale optimization problem, respectively. 3) Greedy Algorithm [21],[22]: the CC-EC scheduling for each user is determined by the user's service request state regardless of other users, and the vBS in slice s corresponding to the user's nearest physical BS is activated if the user requests the service of slice s. Each BS allocates the same BL capacity, transmission power and computing capability to all its served users. Fig. 3 shows the convergence of the RCDDQN-RCDDPG algorithm during the training stage of the proposed E2E-CECN for different numbers of users. We notice that the rewards of E2E-CECN converge after about 1500 training episodes, and the reward of E2E-CECN increases with the increasing number of users.

Fig. 4 plots the E2E throughput of slice 1 and the average service delay of slice 2 versus the numbers of users in slice 1 and slice 2, respectively, for E2E-CECN, E2E-CCN and E2E-ECN. Benefit from the cooperation of CS and ES in CECN, the E2E throughput of slice 1 for E2E-CECN is higher than that for E2E-ECN, and the average service delay of slice 2 for E2E-CECN is lower than that for both E2E-CCN and E2E-ECN. Besides, the data requested by the users are all processed by the CS in E2E-CCN, and more BLC will be consumed during the data transmission procedure as compared with E2E-CECN, while the choices of CS and ES for the data processing have no impact on the access link transmission. According to (3), the E2E throughput of a user is defined as the minimum of its BLC and downlink rate. Hence, the E2E throughput of slice 1 for E2E-CECN approaches that for E2E-CCN.

Figs. 5 and 6 plot the E2E throughput of slice 1 and the average service delay of slice 2 versus the numbers of users in slice 1 and slice 2, respectively, for all the considered algorithms. We observe that a higher E2E throughput of slice 1 and a lower average service delay of slice 2 for E2E-CECN can be obtained by the RCDDQN-RCDDPG algorithm when compared to the DDQN-DDPG algorithm, Twin-Actor DDPG algorithm and Greedy algorithm. This is because more abundant

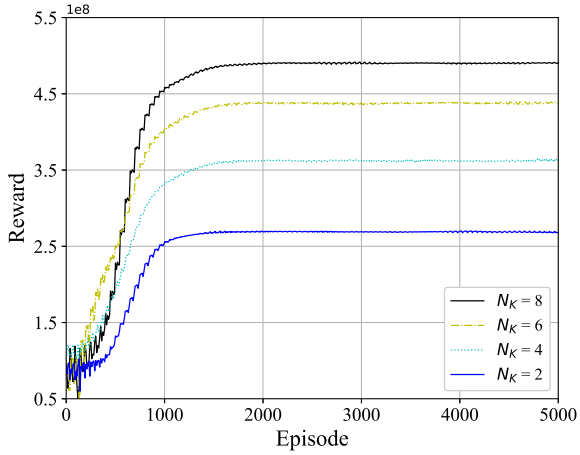


Fig. 3. Convergence of the RCDDQN-RCDDPG algorithm.

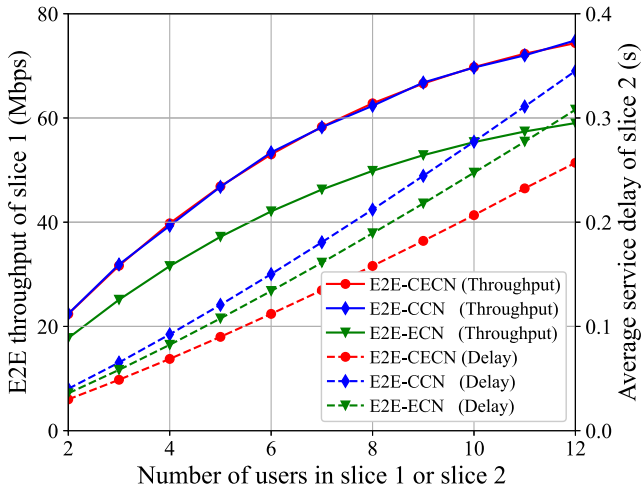


Fig. 4. E2E throughput of slice 1 and average service delay of slice 2 versus the number of users.

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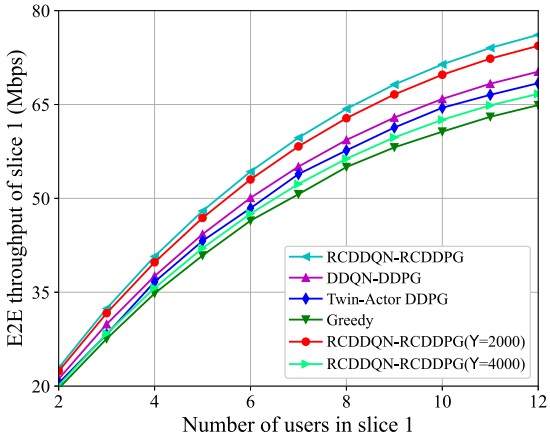


Fig. 5. E2E throughput of slice 1 versus the number of users.

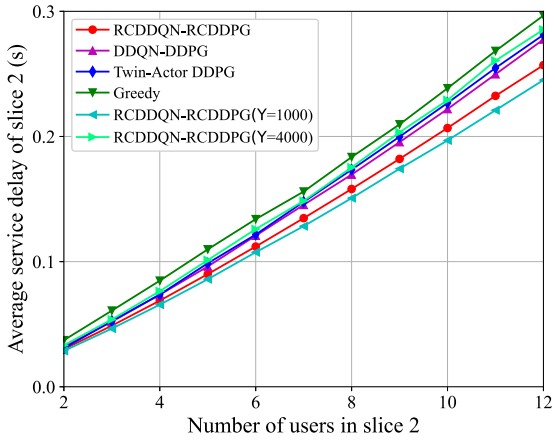


Fig. 6. Average service delay of slice 2 versus the number of users.

CC capability, EC capability and BLC can be allocated to any slice by the RCDDQN-RCDDPG algorithm to better meet its throughput or delay requirements upon the dynamic avenue weights as compared with the fixed avenue weights, and the even resource allocation in the Greedy algorithm can hardly accommodate the variation of the dynamic channel and users' service request states. In addition, with a smaller epoch length Υ for the RCDDQN-RCDDPG algorithm, the vBS-A and CC-EC scheduling can be adjusted more flexibly according to the users' requirements variation. Therefore, the E2E throughput of slice 1 is higher and the average service delay of slice 2 is lower when Υ is smaller as shown in Figs. 5 and 6, respectively.

Fig. 7 plots the BLC and power consumption of the entire physical network versus the number of users for E2E-CECN, E2E-CCN and E2E-ECN. With the mutual cooperation and flexible choices between the CS and ES, less BLC and powers are consumed by the proposed E2E-CECN as compared with E2E-CCN as we can see in Fig. 5. In E2E-ECN, the data requested by the users are all processed by the ES, which means that less BLC will be consumed by E2E-ECN during the data transmission procedure than that by E2E-CECN, and the computing capability of the ES is much lower than that of the CS. Therefore, the BLC and power consumed by E2E-CECN is higher than that by E2E-ECN.

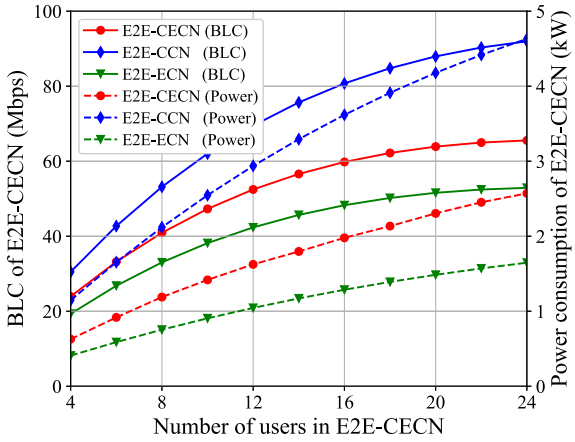


Fig. 7. BLC and Power consumption versus the number of users.

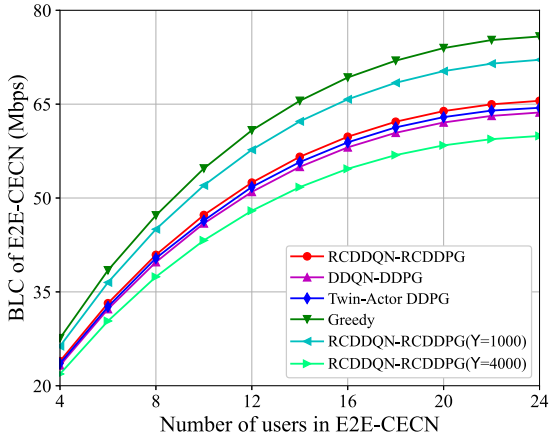


Fig. 8. BLC versus the number of users.

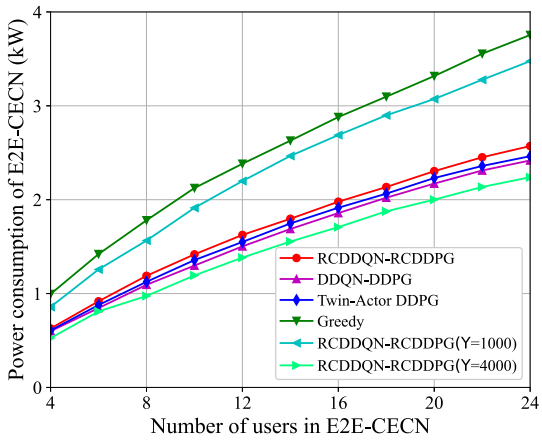


Fig. 9. Power consumption versus the number of users.

Figs. 8 and 9 plot the BLC and power consumption of the entire physical network versus the number of users in E2E-CECN, respectively. Compared with the DDQN-DDPG algorithm and Twin-Actor algorithm, more BLC and power are consumed with the RCDDQN-RCDDPG algorithm as it sacrifices from improving the throughput and delay for E2E-CECN with the dynamic revenue weights. Due to the vBS-A and CC-EC scheduling in the Greedy algorithm considers only the service request state of the

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TABLE III
EXPERIMENTAL RESULTS

Number of users in slice 1 or 2	E2E throughput of slice 1 (Mbps) (RCDDQN-RCDDPG)	Average service delay of slice 2 (s) (RCDDQN-RCDDPG)	E2E throughput of slice 1 (Mbps) (DDQN-DDPG)	Average service delay of slice 2 (s) (DDQN-DDPG)
2	21.787634620119	0.04001278	19.075527676326	0.04595367
3	30.553739212157	0.05882259	27.236705522780	0.06501756
4	38.882246786882	0.07882894	34.614651318371	0.09728564
5	45.573228846825	0.09999207	40.445436769666	0.11589642
6	51.819016510684	0.13203897	46.160345937717	0.16532461
7	56.404759414154	0.15872738	50.556537276539	0.18996482
8	60.508945950641	0.18203376	55.327695781285	0.22437595

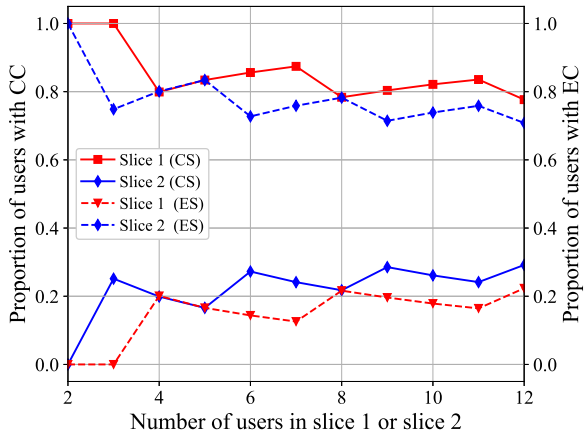


Fig. 10. Proportions of users with CC and EC versus the number of users.

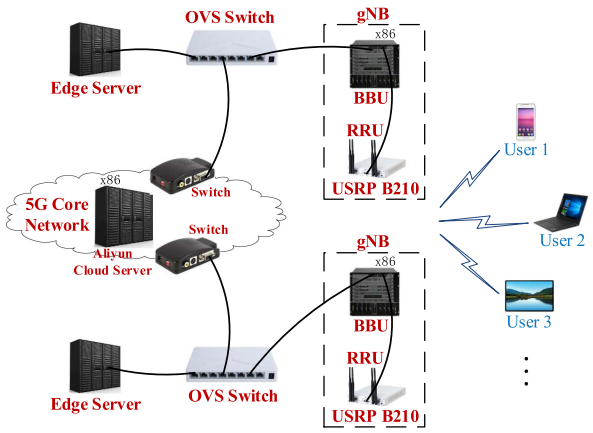


Fig. 11. Testbed for CECN.

individual user instead of the entire network, more vBSs may be activated as compared to the RCDDQN-RCDDPG algorithm, which leads to the higher BLC and power consumption of the Greedy algorithm as shown in Figs. 8 and 9, respectively. Furthermore, we can also notice that the higher BLC and power consumptions are brought by the more flexible vBS-A and CC-EC scheduling under a smaller epoch length Υ than a larger Υ in the RCDDQN-RCDDPG algorithm.

Fig. 10 depicts the proportions of users in E2E-CECN whose requested data are processed by the CS and ES, i.e., $\frac{\sum_{k\in \boldsymbol{N}^s_{\boldsymbol{K}}(t)}\zeta^s_k(T)}{N^s_K(t)}$ and $\frac{\sum_{k\in \boldsymbol{N}^s_{\boldsymbol{K}}(t)}(1-\zeta^s_k(T))}{N^s_K(t)}$, respectively, versus the number of users in slice 1 or slice 2. The CC-EC scheduling indicators $\zeta^s_k(T)$ and numbers of users in any slice $N^s_K(t)$ are collected after our algorithm converges. We find that the data requested by the users in throughput-guaranteed slice 1 tend to be processed by the CS, while the data requested by the users in delay-guaranteed slice 2 tend to be processed by the ES, which conforms to the service advantages of CC and EC, i.e., the CC is more suitable for the services characterized by massive data and intensive computation, while the EC is much more suitable for the delay-sensitive services [1].

The E2E-CECN application scenarios for the high-definition video and smooth video services in a test network as shown in Fig. 11, which is a 3GPP R15-based cellular network built on open-source software and open-air-interface [23]. The 5G core network (CN) is deployed on the Aliyun CS, and Openair-CN-5G is used as the software of the 5G CN. Two gNBs with the operating system of Ubuntu 18.04 connect to the cloud via the Open Virtual Switch (OVS), and a gNB is composed of the Baseband Unit (BBU) and Radio Remote Unit (RRU). A Universal Software Radio Peripheral (USRP) B210 with the frequency band of 70MHz-6GHz is used as the RRU, and Open-Air-Interface-5G is used as the software in the BBU. Both the Aliyun CS and BBU are the general x86 servers. The management and orchestration of the ES is implemented with the assist of Kubernetes. Besides, the OVS MT7621AT with the operating system of OpenWrt R20.7.1 is used as the router to realize the communication among different servers, and Docker is used to realize the virtualization of CECN to create the E2E network slices. Due to lab resource limitations, we test only the E2E throughput and service delay obtained by the RCDDQN-RCDDPG algorithm and DDQN-DDPG algorithm for the number of users varying from 1 to 8in this testbed. The experimental results in Table III show that both the E2E throughput of slice 1 and the average service delay of slice 2 are incremental with the growing number of users. Meanwhile, a higher E2E throughput and lower service delay can be obtained by the RCDDQN-RCDDPG algorithm as compared with the DDQN-DDPG algorithm, which shows the advantage of our proposed RCDDQN-RCDDPG algorithm.

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VI. Conclusion and Future Works

To exploit the synergy between CECN and E2E network slicing to support differentiated services, we optimized E2E-CECN to meet the service requirements of high throughput and low delay E2E network slices. The system-level E2E-CECN utility was defined based on the throughput and delay of the two E2E network slices, respectively, as well as the cost of physical CECN. To reduce the CC/EC switching overhead, the E2E-CECN optimization was formulated into a two-timescale utility maximization problem. The RCDDQN as well as RCDDPG algorithms were proposed to optimize the large timescale joint vBS-A and CC-EC scheduling as well as the small timescale joint BLC, CC/EC capability and transmission power allocations, respectively. Numerical results demonstrated that both the high-throughput and low-delay services can be supported by E2E-CECN with the reasonable BLC and power costs. In our future work, edge caching will be considered in CECN to further enhance the service quality for users.

Appendix A
Proof of NP-Hardness

To prove the hardness of the optimization problem $\mathcal{P}^*$, we first reduce it to a downlink sum-rate maximization problem by assuming that the transmission power is evenly allocated to the users associated with the same base station (BS). The sum-rate maximization problem can be given by

$$\max_{\{\boldsymbol{a}^s(T)\}} \sum_{k\in\boldsymbol{N}^s_{\boldsymbol{K}}(t)} R^s_k(t) = \sum_{k\in\boldsymbol{N}^s_{\boldsymbol{K}}(t)} B_K \log_2\left[1+SINR^s_k(t)\right],$$
$$\text{s.t.} \quad C1: a^s_b(T)\in\{0,1\}, \forall s\in\{1,2\}, b\in\boldsymbol{N}^s_{\boldsymbol{B}}(T), k\in\boldsymbol{N}^s_{\boldsymbol{K}}(t), \tag{26}$$

where the binary vBS association indicator $g^s_{b,k}(t)=1$ indicates that user k in slice s is associated with the nearest vBS b at slot t where $b=\text{argmin}_{b'\in\boldsymbol{N}^s_{\boldsymbol{B}}(T)} r^s_{b',k}$ ($N^s_B=\sum_{b\in\boldsymbol{N}^s_{\boldsymbol{B}}(T)} a^s_b(T)$). Otherwise, $g^s_{b,k}(t)=0$. The number of users associated with vBS b in slice s at slot t is $K^s_b(t)=\sum_{k\in\boldsymbol{N}^s_{\boldsymbol{K}}(t)} g^s_{b,k}(t)$, and the transmission power allocated to user k in slice s at slot t is $P^s_k(t)=\frac{P_B}{K^s_b(t)}$. We can see that the sum-rate depends on the user association with the vBSs, which is affected by the vBS-A. Each user in slice s at slot t is associated with its nearest activated vBS in slice s. Theoretically, any user k ($k\in\boldsymbol{N}^s_{\boldsymbol{K}}(t)$) in slice s can be associated with any vBS b ($b\in\boldsymbol{N}^s_{\boldsymbol{B}}(T)$) if there is no closer activated vBS than vBS b. Therefore, the sum-rate maximization problem is actually a generalized assignment problem (GAP) in which $N^s_K(t)$ users are associated with $N^s_B(T)$ vBSs. The GAP has been proven to be NP-hard in [24], thus the optimization problem $\mathcal{P}^*$ is NP-hard.

Appendix B
Proof of Non-Convexity

As part of the objective function in $\mathcal{P}^*$, the rate of user k in slice s at slot t is given by

$$R^s_k(t)=B_K\log_2\left[1+SINR^s_k(t)\right]. \tag{27}$$

where

$$SINR^s_k(t)=\frac{P^s_k(t)h^s_{b,k}(t)g^s_{b,k}(t)(r^s_{b,k})^{-\alpha}}{\sum_{j\in\boldsymbol{N}^s_{\boldsymbol{K}}(t)\setminus\{k\}}\sum_{d\in\boldsymbol{N}^s_{\boldsymbol{B}}(T)\setminus\{b\}} P^s_j(t)h^s_{d,k}(t)g^s_{d,j}(t)(r^s_{d,k})^{-\alpha}+\sigma^2}.$$

The first derivative of $R^s_k(t)$ with respect to $P^s_k(t)$ can be expressed as:

$$\left(R^s_k(t)\right)'=\frac{d}{d\left(P^s_k(t)\right)}R^s_k(t)=\frac{B_K}{\ln 2}\cdot\frac{\Lambda'(t)}{\Phi'(t)}, \tag{28}$$

where

$$\Lambda'(t)=\frac{h^s_{b,k}(t)g^s_{b,k}(t)(r^s_{b,k})^{-\alpha}}{\sum_{j\in\boldsymbol{N}^s_{\boldsymbol{K}}(t)\setminus\{k\}}\sum_{d\in\boldsymbol{N}^s_{\boldsymbol{B}}(T)\setminus\{b\}} P^s_j(t)h^s_{d,k}(t)g^s_{d,j}(t)(r^s_{d,k})^{-\alpha}+\sigma^2},$$

and

$$\Phi'(t)=1+\frac{P^s_k(t)h^s_{b,k}(t)g^s_{b,k}(t)(r^s_{b,k})^{-\alpha}}{\sum_{j\in\boldsymbol{N}^s_{\boldsymbol{K}}(t)\setminus\{k\}}\sum_{d\in\boldsymbol{N}^s_{\boldsymbol{B}}(T)\setminus\{b\}} P^s_j(t)h^s_{d,k}(t)g^s_{d,j}(t)(r^s_{d,k})^{-\alpha}+\sigma^2}.$$

The second derivative of $R^s_k(t)$ with respect to $P^s_k(t)$ is given by:

$$\left(R^s_k(t)\right)''=\frac{d}{d\left(P^s_k(t)\right)}\left(R^s_k(t)\right)'=-\frac{B_K}{\ln 2}\cdot\frac{\Lambda''(t)}{\Phi''(t)}, \tag{29}$$

where

$$\Lambda''(t)=\left[\frac{h^s_{b,k}(t)g^s_{b,k}(t)(r^s_{b,k})^{-\alpha}}{\sum_{j\in\boldsymbol{N}^s_{\boldsymbol{K}}(t)\setminus\{k\}}\sum_{d\in\boldsymbol{N}^s_{\boldsymbol{B}}(T)\setminus\{b\}} P^s_j(t)h^s_{d,k}(t)g^s_{d,j}(t)(r^s_{d,k})^{-\alpha}+\sigma^2}\right]^2,$$

and

$$\Phi''(t)=\left[1+\frac{P^s_k(t)h^s_{b,k}(t)g^s_{b,k}(t)(r^s_{b,k})^{-\alpha}}{\sum_{j\in N^s_K(t)\setminus\{k\}}\sum_{d\in\boldsymbol{N}^s_{\boldsymbol{B}}(T)\setminus\{b\}} P^s_j(t)h^s_{d,k}(t)g^s_{d,j}(t)(r^s_{d,k})^{-\alpha}+\sigma^2}\right]^2.$$

Both $\Lambda''(t)$ and $\Phi''(t)$ are positive, and $\left(R^s_k(t)\right)''$ is negative. Hence, $R^s_k(t)$ is concave in $P^s_k(t)$, and the non-convexity of the optimization problem $\mathcal{P}^*$ is proven.

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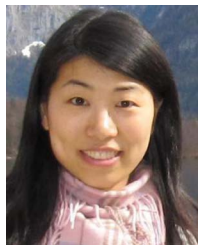
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